



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 21/07/2016*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for $n=1, \dots, N$, where A is the constant signal of interest (SOI), modelled as a random variable uniformly distributed between 0 and 1, and $\{W[n]; n=1, \dots, N\}$ are independent random variables modeling additive noise, which is independent of A . Each noise sample is Gaussian distributed, zero mean and with variance σ^2 .

- (1) Derive the Linear Minimum Mean Square Error (LMMSE) estimator of the random parameter A given the observed data vector $\mathbf{X} \triangleq [X[1] \ X[2] \ \dots \ X[N]]^T$ and express it as a function of the parameter γ defined as follows: $\gamma \triangleq \text{var}\{A\} / \text{var}\{W[n]\} = \text{var}\{A\} / \sigma^2$.
- (2) Calculate its mean square error (MSE) and express it as a function of the parameter γ .
- (3) Can we state that the MMSE estimator and the LMMSE estimator coincide in this case?
- (4) Determine numerically the LMMSE and its MSE when $N=10$ and $\sigma^2=1$.

Woodbury's identity: $(\mathbf{I} + \gamma \mathbf{1} \mathbf{1}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1 + \gamma \mathbf{1}^T \mathbf{1}} \mathbf{1} \mathbf{1}^T = \mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{1} \mathbf{1}^T$



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Solution:

$$\mathbf{X} \triangleq [X[1] \ X[2] \ \cdots \ X[N]]^T, \quad E\{A\} = \frac{1}{2}, \quad E\{X[n]\} = E\{A\} = \frac{1}{2} \quad \Rightarrow \quad E\{\mathbf{X}\} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \vdots \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2} \mathbf{1}$$

$$\hat{A}_{LMMSE} = \sum_{n=1}^N a_n X[n] + b = \mathbf{a}^T \mathbf{X} + b, \quad \varepsilon = A - \hat{A}_{LMMSE} = A - \mathbf{a}^T \mathbf{X} - b$$

Unbiasedness:

$$E\{\varepsilon\} = E\{A\} - \mathbf{a}^T E\{\mathbf{X}\} - b = 0 \quad \Rightarrow \quad b = E\{A\} - \mathbf{a}^T E\{\mathbf{X}\} = \frac{1}{2} - \frac{1}{2} \mathbf{a}^T \mathbf{1}$$

Orthogonality Principle:

$$\varepsilon = A - \mathbf{a}^T \mathbf{X} - \left(\frac{1}{2} - \frac{1}{2} \mathbf{a}^T \mathbf{1} \right) = \left(A - \frac{1}{2} \right) - \mathbf{a}^T \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)$$

$$E\{\varepsilon \mathbf{X}^T\} = 0 \quad \Rightarrow \quad E\left\{ \varepsilon \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)^T \right\} = 0$$

$$E\left\{ \varepsilon \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)^T \right\} = E\left\{ \left(A - \frac{1}{2} \right) \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)^T \right\} - \mathbf{a}^T E\left\{ \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right) \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)^T \right\} = \mathbf{c}_{AX} - \mathbf{a}^T \mathbf{C}_X = 0$$

$$\Rightarrow \quad \mathbf{a}^T = \mathbf{c}_{AX} \mathbf{C}_X^{-1} \quad \Rightarrow \quad \mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_{AX}^T = \mathbf{C}_X^{-1} \mathbf{c}_{XA}$$

$$\mathbf{c}_{AX} \triangleq E\left\{ \left(A - \frac{1}{2} \right) \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right)^T \right\} = \mathbf{c}_{XA}^T$$

$$\mathbf{c}_{XA} \triangleq E\left\{ \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right) \left(A - \frac{1}{2} \right) \right\} = \text{cov}\{X[n], A\} \mathbf{1} = \text{var}\{A\} \mathbf{1} = \frac{1}{12} \mathbf{1}$$



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$$\mathbf{C}_X = \mathbf{C}_A + \mathbf{C}_W = \text{var}\{A\} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} + \begin{bmatrix} \text{var}\{W[1]\} & 0 & \cdots & 0 \\ 0 & \text{var}\{W[2]\} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{var}\{W[N]\} \end{bmatrix}$$

$$= \frac{1}{12} \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I} = \sigma^2 (\gamma \mathbf{1}\mathbf{1}^T + \mathbf{I})$$

$$\mathbf{C}_X^{-1} = \frac{1}{\sigma^2} (\gamma \mathbf{1}\mathbf{1}^T + \mathbf{I})^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{1}\mathbf{1}^T \right)$$

$$\mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_{XA} = \frac{1}{\sigma^2} \left(\mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{1}\mathbf{1}^T \right) \frac{1}{12} \mathbf{1} = \gamma \left(\mathbf{1} - \frac{\gamma}{1 + N\gamma} \mathbf{1}\mathbf{1}^T \mathbf{1} \right) = \gamma \left(1 - \frac{N\gamma}{1 + N\gamma} \right) \mathbf{1}$$

$$= \frac{\gamma}{1 + N\gamma} \mathbf{1} = \frac{N\gamma}{1 + N\gamma} \cdot \frac{1}{N} \mathbf{1} = \alpha \frac{1}{N} \mathbf{1}, \quad \text{where: } \alpha \triangleq \frac{N\gamma}{1 + N\gamma}$$

$$b = \frac{1}{2} (1 - \mathbf{a}^T \mathbf{1}) = \frac{1}{2} \left(1 - \frac{\gamma}{1 + N\gamma} \mathbf{1}^T \mathbf{1} \right) = \frac{1}{2} \left(1 - \frac{N\gamma}{1 + N\gamma} \right) = \frac{1}{2} \cdot (1 - \alpha)$$

$$\hat{A}_{LMMSE} = \alpha \frac{1}{N} \mathbf{1}^T \mathbf{X} + \frac{1}{2} \cdot (1 - \alpha) = \alpha \hat{A}_{SM} + (1 - \alpha) \eta_A$$

where $\hat{A}_{SM} \triangleq \frac{1}{N} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \sum_{n=1}^N X[n]$ is the Sample Mean estimator

$$\begin{aligned} \text{MSE} \{ \hat{A}_{LMMSE} \} &= E \{ \varepsilon^2 \} = E \{ \varepsilon A \} = E \left\{ \varepsilon \left(A - \frac{1}{2} \right) \right\} \\ &= E \left\{ \left(A - \alpha \frac{1}{N} \mathbf{1}^T \mathbf{X} - \frac{1}{2} \cdot (1 - \alpha) \right) \left(A - \frac{1}{2} \right) \right\} = E \left\{ \left(A - \frac{1}{2} - \alpha \frac{1}{N} \mathbf{1}^T \mathbf{X} - \alpha \frac{1}{2} \right) \left(A - \frac{1}{2} \right) \right\} \\ &= E \left\{ \left(A - \frac{1}{2} - \alpha \frac{1}{N} \mathbf{1}^T \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right) \right) \left(A - \frac{1}{2} \right) \right\} = E \left\{ \left(A - \frac{1}{2} \right)^2 \right\} - \alpha \frac{1}{N} \mathbf{1}^T E \left\{ \left(\mathbf{X} - \frac{1}{2} \mathbf{1} \right) \left(A - \frac{1}{2} \right) \right\} \\ &= \text{var}\{A\} - \frac{1}{N} \mathbf{1}^T \mathbf{c}_{XA} = \frac{1}{12} - \frac{\alpha}{N} \cdot \frac{1}{12} \mathbf{1}^T \mathbf{1} = \frac{1}{12} (1 - \alpha) = \frac{1}{12} \cdot \frac{1}{1 + N\gamma} \end{aligned}$$



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$$\text{For } N=10 \text{ and } \sigma^2=1 \Rightarrow \gamma=\frac{1}{12} \Rightarrow \alpha=\frac{10/12}{1+10/12}=\frac{10}{22}=0.4545$$

$$\hat{A}_{LMMSE}=0.04545 \cdot \mathbf{1}^T \mathbf{X} + \frac{1}{2} \cdot (1-0.4545) = 0.04545 \sum_{n=1}^{10} X[n] + 0.2727$$

$$MSE\{\hat{A}_{LMMSE}\}=\frac{1}{12} \cdot \frac{1}{1+N\gamma}=\frac{1}{12} \cdot \frac{1}{1+10 \cdot \frac{1}{12}}=\frac{1}{22}=0.04545$$